MAST20030 Differential Equations
— Problem Sheet 2: Power Series —

1. Consider the following ODEs:
   (a) $y''(x) - 4y(x) = 0$ (b) $y''(x) + 4y(x) = 0$

   For each ODE:

   (i) Find the general solution using a power series solution around $x = 0$.
   (ii) Using the Weierstrass test, prove that the series solutions are uniformly convergent.
   (iii) Find the general solution using the methods from Topic 1.
   (iv) Find the Taylor series representation of your solutions in (iii) to verify your solutions in (i) are correct.

2. Use power series about $t = 0$ to find as many solutions as you can to the following ODEs. If possible, give a closed form solution for each and determine the radius of convergence. If not, give the first six non-zero terms of the series.

   (a) $(1-t)y''(t) + y(t) = 0$
   (b) $(2+t^2)y''(t) - ty'(t) + 4y(t) = 0$
   (c) $ty''(t) + y'(t) + ty(t) = 0$
   (d) $(1+t^2)y''(t) - 4ty'(t) + 6y(t) = 0$
   (e) $(4-t^2)y''(t) + 2y(t) = 0$
   (f) $2y''(t) + ty'(t) + 3y(t) = 0$
   (g) $y''(t) + ty'(t) + y(t) = 0$
   (h) $y''(t) - ty'(t) - t^2y(t) = 0$
   (i) $ty''(t) + 2y'(t) + ty(t) = 0$
   (j) $ty''(t) + 3y'(t) - y(t) = 0$

3. Consider the function $f\colon \mathbb{R} \to \mathbb{R}$ defined by $f(t) = \begin{cases} e^{-\frac{1}{t^2}} & t \neq 0 \\ 0 & t = 0 \end{cases}$.

   (a) Show that f is continuous at $t = 0$.
   (b) Show that f is differentiable at $t = 0$ and compute $f'(0)$.
   (c) Show that f is twice differentiable at $t = 0$ and compute $f''(0)$.
   (d) Show that f is thrice differentiable at $t = 0$ and compute $f'''(0)$.
   (e) Using your insight from (a)-(c), argue that f is j-times differentiable at $t = 0$, for all $j = 0, 1, 2, \ldots$, and give the value of $f^{(j)}(0)$. [For an extra (non-examinable) challenge, prove this using induction.]
   (f) Show that the Taylor series of f converges for all t, but that it does not converge to $f(t)$.

4. Try to find a general power series solution about $t = 0$ for

$$2t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 3t\frac{\mathrm{d}y}{\mathrm{d}t} - (1+t)y = 0.$$

   What goes wrong? Use the form of a generalised power series, $t^\lambda \sum_{j=0}^{\infty} c_j t^j$ to find the solution.

5. In lectures, we studied an example of a Bessel equation $ty''(t) + y'(t) + ty(t) = 0$ and showed that it has only one power series solution about $t = 0$ (up to multiplying by a constant):

$$y_1(t) = \sum_{m=0}^{\infty} \frac{(-1)^m t^{2m}}{4^m (m!)^2}.$$

Moreover, the radius of convergence was shown to be infinite.

(a) Compute the wronskian of y_1 and another linearly independent solution y_2 and use this to show that y_2 diverges like $\log t$ for $t \to 0^+$. [We checked this using reduction of order in lectures, it's really the same!]

(b) Your wronskian analysis should naturally lead you to a generalised series expansion for $y_2(t)$ about $t = 0$. Find the first three non-vanishing terms in this series.

(c) Explain why this analysis suggests that y_2 may be expressed as a series of the form

$$y_2(t) = \log t \, y_1(t) + \sum_{j=0}^{\infty} c_j t^j.$$

Compute the c_j and determine the t for which $y_2(t)$ converges.

6. Uniform convergence is often introduced because of the following nice property:

If a sequence $w_n(x)$ of *continuous* functions converges uniformly to $w(x)$ on a domain D, then $w(x)$ is also continuous on D.

(a) Give an example of a sequence $w_n(x)$ of continuous functions that converges *pointwise*, i.e., at each x, in a domain D to a discontinuous function $w(x)$.

(b) Consider the series

$$\sum_{j=0}^{\infty} \frac{x^2}{(1+x^2)^j}.$$

Show that this series converges for all x but that the convergence is not uniform. *Hint: you can easily evaluate this series explicitly, but you need to be a little careful.*

(c) Consider $w_n(x) = \sqrt{x^2 + \frac{1}{n^2}}$. As $n \to \infty$, show that $w_n(x)$ converges *uniformly* to $w(x) = |x|$ on $\mathbb{R}$. We have noted above that uniform convergence preserves continuity. Does it preserve differentiability?

MAST20030 Differential Equations
— Solutions to Problem Sheet 2: Power Series —

1. (a) (i) $C_1 \displaystyle\sum_{n=0}^{\infty} \frac{4^n}{(2n)!} t^{2n} + C_2 \sum_{n=0}^{\infty} \frac{2^{2n+1}}{(2n+1)!} t^{2n+1}$

   (ii) Choose an $L \in \mathbb{R}$ where $0 < L$. Substitute L into the series, show that for $t \in [-L, L]$ this term gives the maximum error and that the resulting series converges.

   (iii) $C_1 \cosh 2t + C_2 \sinh 2t$.

   (iv) Taylor series as in (i).

   (b) (i) $C_1 \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n 4^n}{(2n)!} t^{2n} + C_2 \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n+1}}{(2n+1)!} t^{2n+1}$

   (ii) Choose an $L \in \mathbb{R}$ where $0 < L$. Substitute L into the series, remove the $(-1)^n$ term, show that for $t \in [-L, L]$ this term gives the maximum error and that the resulting series converges.

   (iii) $C_1 \cos 2t + C_2 \sin 2t$.

   (iv) Taylor series as in (i).

2. (a) $1 - \dfrac{t^2}{2} - \dfrac{t^3}{6} - \dfrac{t^4}{24} - \dfrac{t^5}{60} - \dfrac{7t^6}{720}$ and $t - \dfrac{t^3}{6} - \dfrac{t^4}{12} - \dfrac{t^5}{24} - \dfrac{t^6}{40} - \dfrac{17t^7}{1008}$
   (radius of convergence is 1 for both, but this is hard to show!)

   (b) $1 - t^2 + \dfrac{t^4}{6} - \dfrac{t^6}{30} + \dfrac{t^8}{120} - \dfrac{13t^{10}}{5400}$ and $t - \dfrac{t^3}{4} + \dfrac{7t^5}{160} - \dfrac{19t^7}{1920} + \dfrac{247t^9}{92\,160} - \dfrac{16549t^{11}}{20\,275\,200}$
   (radius of convergence is $\sqrt{2}$ for both)

   (c) Only one power series solution: $1 - \dfrac{t^2}{4} + \dfrac{t^4}{64} - \dfrac{t^6}{2304} + \dfrac{t^8}{147\,456} - \dfrac{t^{10}}{14\,745\,600}$
   (radius of convergence is ∞)

   (d) Two closed form solutions: $1 - 3t^2$ and $t - \dfrac{t^3}{3}$

   (e) One closed form solution: $1 - \dfrac{t^2}{4}$ and one power series solution:
   $t - \dfrac{t^3}{12} - \dfrac{t^5}{240} - \dfrac{t^7}{2240} - \dfrac{t^9}{16\,128} - \dfrac{t^{11}}{101\,376}$
   (the power series turns out to be $\frac{t}{2} - \frac{1}{4}(t^2 - 4)\operatorname{arctanh}(\frac{t}{2})$, with radius of convergence 2)

   (f) $1 - \dfrac{3t^2}{4} + \dfrac{5t^4}{32} - \dfrac{7t^6}{384} + \dfrac{3t^8}{2048} - \dfrac{11t^{10}}{122\,880}$ and $t - \dfrac{t^3}{3} + \dfrac{t^5}{20} - \dfrac{t^7}{210} + \dfrac{t^9}{3024} - \dfrac{t^{11}}{55\,440}$
   (radius of convergence is ∞ for both; the series with even powers is $-\frac{1}{2}(t^2 - 2)e^{-t^2/4}$)

   (g) $1 - \dfrac{t^2}{2} + \dfrac{t^4}{8} - \dfrac{t^6}{48} + \dfrac{t^8}{384} - \dfrac{t^{10}}{3840}$ and $t - \dfrac{t^3}{3} + \dfrac{t^5}{15} - \dfrac{t^7}{105} + \dfrac{t^9}{945} - \dfrac{t^{11}}{10\,395}$
   (radius of convergence is ∞ for both; the series with even powers is $e^{-t^2/2}$)

   (h) $1 + \dfrac{t^4}{12} + \dfrac{t^6}{90} + \dfrac{3t^8}{1120} + \dfrac{41t^{10}}{113\,400} + \dfrac{571t^{12}}{11\,975\,040}$ and $t + \dfrac{t^3}{6} + \dfrac{3t^5}{40} + \dfrac{13t^7}{1008} + \dfrac{119t^9}{51\,840} + \dfrac{41t^{11}}{134\,400}$
   (radius of convergence is ∞ for both (hard to show!))

(i) Only one power series solution: $1 - \dfrac{t^2}{6} + \dfrac{t^4}{120} - \dfrac{t^6}{5040} + \dfrac{t^8}{362\,880} - \dfrac{t^{10}}{39\,916\,800}$
(the power series is $\frac{\sin t}{t}$ with radius of convergence ∞)

(j) Only one power series solution: $1 + \dfrac{t}{3} + \dfrac{t^2}{24} + \dfrac{t^3}{360} + \dfrac{t^4}{8640} + \dfrac{t^5}{302\,400}$
(radius of convergence is ∞)

3. (f) Taylor series converges to zero for all t.

4. Substitute generalised series into ODE. Examining the $n = 0$ term, we have non-trivial solutions when $\lambda = -1$ and $\lambda = \dfrac{1}{2}$. The general solution is

$$y(t) = C_1 t^{\frac{1}{2}} \sum_{n=0}^{\infty} \frac{2^n}{(3+2n)(2n+1)!} t^n + \frac{C_2}{t}\left(1 - \sum_{n=1}^{\infty} \frac{2^{n-1}}{n(2n-2)!} t^n\right)$$

5. (a) Proof follows the analysis performed in lectures

(b) Our Wronskian analysis leads to $y_2(t) = y_1(t) \displaystyle\int \frac{1}{t y_1^2(t)} \mathrm{d}t$. Series expanding the integrand gives $\dfrac{1}{t y_1^2(t)} = \dfrac{1}{t} + \dfrac{t}{2} + \dfrac{5t^3}{32} + O(t^4)$. Upon integrating wrt t and multiplying by $y_1(t)$ we get $y_2(t) = y_1(t)\left(\log t + \dfrac{t^2}{4} + \dfrac{5t^4}{128} + O(t^5)\right)$.

(c) The only term in (b) which cannot be represented by a usual power series is $\log(t) y_1(t)$, suggesting the form provided in the question. Substituting into the ODE eventually leads to the recurrence relation $c_{2m} = -\dfrac{1}{4m^2} c_{2m-2} - \dfrac{(-1)^m}{m 4^m (m!)^2}$ with odd $c_{2n-1} = 0$. The solution will converge for $t > 0$.

6. (a) There are many such sequences. In lectures, we introduced uniform convergence with the example $w_n(x) = x^n$ on $D = [0, 1]$. This sequence converges pointwise to

$$w(x) = \begin{cases} 1 & \text{if} \quad x = 1, \\ 0 & \text{otherwise} \end{cases}$$

which is discontinuous, as required.

(b) By making the substitution $u = \frac{1}{1+x^2}$, we see that this is just a geometric series in disguise:

$$\sum_{j=0}^{\infty} \frac{x^2}{(1+x^2)^j} = x^2 \sum_{j=0}^{\infty} u^j = \frac{x^2}{1-u} = \frac{x^2}{1 - \frac{1}{1+x^2}} = 1 + x^2, \qquad |u| < 1.$$

But, $|u| < 1$ means $1 + x^2 > 1$, i.e., $x \neq 0$. So for $x \neq 0$, this series converges to $1 + x^2$. However, for $x = 0$, this series converges to 0 because each term is 0. The series thus converges for all x, but it converges (pointwise) to a discontinuous function. Therefore, the convergence cannot be uniform.

(c) We have $w_n(x) \to w(x)$ pointwise as $n \to \infty$. To show uniform convergence we have to show that the maximum error tends to 0. This error is

$$E_n(x) = |w_n(x) - w(x)| = \sqrt{x^2 + \frac{1}{n^2}} - \sqrt{x^2}$$

$$\Rightarrow \qquad E_n'(x) = \frac{x}{\sqrt{x^2 + \frac{1}{n^2}}} - \frac{x}{\sqrt{x^2}} = \begin{cases} \frac{x}{\sqrt{x^2+\frac{1}{n^2}}} + 1 & if \text{x¡0,} \\ \frac{x}{\sqrt{x^2+\frac{1}{n^2}}} - 1 & if \text{x¿0.} \end{cases}$$

The derivative is undefined at $x = 0$. As the derivative is never 0, the maximum either occurs at $x = 0$ or in the limit as $x \to \pm\infty$. But $E_n(x) \to 0$ as $x \to \pm\infty$, while $E_n(0) = \frac{1}{n}$. The *maximum* error is thus $\frac{1}{n}$ which does indeed tend to 0 as $n \to \infty$, proving that $w_n(x) \to w(x)$ uniformly on $\mathbb{R}$. However, the $w_n(x)$ are differentiable while $w(x) = |x|$ is not (at $x = 0$). We conclude that uniform convergence need not preserve differentiability.